Adult Guide — E02 One Whole, Many Parts

Review version 1.0.0 Author: Maria Smith Audience: parents, carers, early-years practitioners and teachers Final-publication status: awaiting Maria Smith's approval

Purpose

The child rejoins Mia, Sol and Tavi immediately after E01. The parcel contains one Moon Lantern and a picture card asking the team to take it to the balcony. A small door blocks the way, and the lantern is too wide to fit through it. The book then holds one simple question in view: How can the whole lantern get through the small door?

Pax gives the child a four-step plan: check the whole, choose and count every part, carry every part through, then rebuild the same whole. Every activity is one necessary part of that plan. The vocabulary follows the experience rather than replacing it. The completed lantern reaches the balcony and naturally begins the two-label pattern that previews E03.

Opening recap and invitation

Book Two and Level Two begin with a short story recap so a child understands why the parcel is waiting and what the team is doing next. The recap must also work for a child who starts here instead of completing Level One.

The book opening and the game's non-playable prelude should:

1. welcome the child back to Mia, Sol and Tavi, or introduce the trio in one sentence for a new child;
2. explain that the team followed five clues, opened the Star Door and learned to describe exactly what they found instead of inventing “nothing”;
3. show that the parcel arrived at the end of that mystery and is now waiting beside the filed map;
4. let the team open the parcel and reveal the complete Moon Lantern before it is moved or taken apart;
5. show the picture card asking for the lantern at the balcony and the small door that blocks the whole lantern; and
6. invite the child to discover how one whole can become several parts, how to count every part once, and how the same whole can be built again—without revealing the plan's answers.

The prelude is a warm continuation, not a spoken contents list. On replay it may be watched again or skipped, but skipping it must not mark any activity as complete.

The permanent cast and Pax

Mia, Sol and Tavi remain the main trio. E02 introduces exactly one new guest: Pax, the patient piece-keeper. Pax checks fitted trays and asks process questions but does not identify the correct layout or count before an attempt.

Pax may return in a later book only when components, partitions or reassembly are relevant. A later callback may ask whether every place is filled or whether pieces overlap; it must not reveal a new answer.

Narrative cause chain

1. The E01 parcel supplies the Moon Lantern and picture card.
2. The small doorway creates the one practical problem.
3. The child identifies the original whole so they know what must be rebuilt.
4. The child chooses four complete, separate, same-size parts.
5. The child checks each carrying spot once so no part is missed or repeated.
6. All four parts pass through the doorway.
7. The child compares visible sizes and checks held and whole counts.
8. The child fills the gap and keeps the unrelated extra outside.
9. Every lantern part returns to the round frame; the tray is empty.
10. The child joins four separate one-part counts and chooses the exact total.
11. The four counted parts are fitted together as one whole lantern.

If the child loses the thread, point to the small door and say: “The whole lantern will not fit. What is our plan?” Then use the four short actions: “Check it. Take it apart. Carry every part. Build it again.”

Experience-before-vocabulary

- whole — all of this registered thing, with none of it missing and nothing extra added;
- count — keep the complete generated row by visiting every item once, skipping none and repeating none;
- complete partition — all of the agreed whole is covered by its registered pieces;
- separate / non-overlapping — one piece does not sit over another piece;
- equal pieces — pieces made the same size in this particular partition;
- unequal pieces — visible pieces that are not the same size;
- held part — one or more chosen pieces contained in the complete partition;
- held count — how many registered pieces are chosen; and
- whole count — how many registered pieces make the complete partition;
- plus / addition — join separate counted groups while keeping every source item exactly once; and
- equals — both sides name the same exact total, here the same four visible lantern parts.

Do not introduce parameter-free coordinate, fibre, candidate grammar, multiplicity or depth-independent closure to the child. Those exact terms remain here to protect the source boundary.

Exact SFT source and dependency order

1. Structural One

Claim: `SFT-FOUNDATION-ONE-001` Dependency: `SFT-ROOT-THERE-IS-NO-NOTHING` Status: independently replicated Provenance: forward forcing Closure: depth-independent Receipt: `sha256:68332624c276dc5d0ddfd568a26b95e20d7d5665fcdb7e825ff1b7b19bf1d51c`

> The unique minimal representation of the admitted root occurrence is its complete self-whole with no unforced addition; this structural identity is the One.

The complete generated product classifies root coverage as none, proper or complete, crossed with absence or presence of unforced extra material. Six forms are decided. Only `exact-self-whole` retains the complete admitted occurrence without omission or addition. Arithmetic and fractions are not imported into this claim.

2. Exact positive finite count

Claim: `SFT-FOUNDATION-COUNT-001` Dependency: `SFT-FOUNDATION-ONE-001` Status: independently replicated Provenance: forward forcing Closure: depth-independent across every generated nonempty finite extension Receipt: `sha256:4b2411716cbef2fb9fa7aac58b24ce5eb3c0c59fe7230fd703fd89b5260f6e23`

> Every generated nonempty finite succession of structural-One occurrences has one exact positive count representation: its complete generation trace, retaining every generated occurrence once and adding none.

The grammar has ten representation classes across coverage, retained multiplicity and ungenerated extra occurrences. Only `complete-coverage__once__no-extra` survives. E02 translates omission, duplication and addition into a four-light bridge. The bridge is a demonstration, not the formal proof.

3. Exact positive part coordinate

Claim: `SFT-FOUNDATION-PART-001` Dependency: `SFT-FOUNDATION-COUNT-001` Status: independently replicated Provenance: forward forcing Closure: depth-independent within the registered positive finite partition boundary Receipt: `sha256:87b40affae50458ba5492af91778fbca79a8de0bc3b951e520535093c2b7b728`

> For every registered finite partition of the structural One that is complete, disjoint and equal-fibred, and every nonempty held selection contained within it, the unique parameter-free exact part coordinate is the pair of positive counts identifying the held selection and the complete partition.

The formal grammar contains 192 generated alternatives. It classifies whole coverage, overlap, fibre equality, selection position, held-count presence, whole-count presence and extra coordinates. Exactly `complete__disjoint__equal__within__held-present__whole-present__no-extra` survives.

4. Exact arithmetic kernel

Claim: `SFT-MATH-EXACT-ARITHMETIC-001` Dependencies: exact positive count, exact part coordinate, part equivalence and form enforcement Status: independently replicated Provenance: forward forcing Closure: depth-independent Receipt: `sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418`

This downstream result admits exact arithmetic only after the generative count and part structures are sealed. Conventional symbols are correspondence labels after that derivation; they do not supply its authority.

5. Junction addition

Claim: `SFT-MATH-ARITH-JUNCTION-ADDITION-002` Dependencies: exact arithmetic, form enforcement and generated succession Status: independently replicated Provenance: forward forcing Closure: depth-independent Receipt: `sha256:ecbb32a37fb10022315429bb39d88f49437fc856396d0b8db42fca49b2032493`

> Addition is the complete junction of two held-disjoint generated traces,
>
> with every source unit retained exactly once.

E02 repeats that lawful junction to make the child-facing equation `1 + 1 + 1 + 1 = 4 parts`. The four source parts remain visible and are neither lost nor repeated. The later reassembly statement is deliberately written `4 parts → 1 whole lantern`, not `4 = 1`, because parts and whole lantern name different units.

The non-negotiable boundary

E02 does not derive or teach as SFT authority:

- numerical zero or an empty held selection;
- negative, imaginary, irrational, decimal or floating proof quantity;
- completed infinity or an ungenerated continuum;
- division imported as a foundational operation;
- fractions or the claim that each pictured part is a numerical quarter;
- equivalence between differently generated partitions; or
- the claim that every object called "one piece" has equal size.

The child-facing label `2 HELD · 4 MAKE THE WHOLE` keeps the exact positive pair. Do not silently replace it with a decimal. Do not claim it is interchangeable with `1 held · 2 make the whole`; cross-partition equivalence belongs to the separate admitted downstream theorem and is deliberately postponed.

How to share each challenge/reveal pair

1. Read the challenge and pause.
2. Let the child look across every pictured alternative.
3. Accept pointing, gaze, sign, speech, movement or communication aids.
4. Ask one process question without naming the answer.
5. Turn to the reveal and compare the same objects in the same positions.

6. Keep a first attempt that differed; identify exactly what changed.
7. Replay with safe large objects if useful.

The reveal is a shared check, not a score screen.

Page guidance and answers

Pages 3–8 — parcel, problem and plan

Pages 3–6 establish the previous-book callback, the one clear problem, the child-facing question, the four-step plan and the single new guest. Page 7 answer: the complete round lantern, not the lantern with a gap and not the complete lantern plus an attached extra. Page 8 introduces whole only after the child has compared all three.

Pages 9–13 — four carrying spots

Page 10 asks the child to visit the four visible spots once each. Page 12 preserves Sol's check `1, 2, 3, 3, 4`; the problem is the repeated third spot. Page 13 explicitly keeps that failed attempt and rebuilds the trace.

Pages 14–17 — choose the four-part plan

The accepted layout is the one that fills the entire agreed outline, has no overlap and uses same-sized pieces. A layout can fail more than one condition. Ask all three questions rather than accepting a familiar-looking picture. Pages 16–17 add the observable symmetry check only after the child has looked: left and right match around the middle, and top and bottom match around the middle. Symmetrical names that balanced match; it is not used as an unexplained instruction.

Pages 18–20 — carry every part

Every lantern part passes through the same small doorway. One carrying spot is used for each part so the child can see that none was missed or repeated.

Pages 21–22 — check the plan's sizes

Pair A is equal-sized; Pair B is visibly unequal. Unequal pieces are not called false or bad. The narrower claim is that counts alone cannot identify equal portions when the partition pieces are unequal.

Pages 23–25 — practice-frame check: gap and extra

The flat frame is a checking picture, not the final lantern rebuild. Page 23 is not complete because one registered place is unfilled. Page 24 returns the registered fourth part, then explicitly returns all four separate parts to the carrying tray. Page 25 keeps the extra triangle visible and outside; it does not disappear and is not absorbed into the lantern by a word. This keeps the story state coherent for the addition activity on pages 28–29 and the one final rebuild on page 30.

Pages 26–27 — held and whole counts

Answer: two pieces are held and four pieces make the complete registered partition. Both positive counts stay visible. Do not extend the activity to an empty held selection in this book.

Pages 28–29 — exact addition and equality

Answer: four parts. Invite the child to touch each separate part once before choosing the total. Read + as “plus” and explain that it joins the separate counts. Read = as “equals” and explain that both sides count the same four parts. Do not call the pieces quarters and do not hide or merge them while the equation is being checked.

Pages 30–32 — answer the story question

Page 30 changes from counting to reassembly: the exact four parts are fitted together as the same whole lantern. The arrow is a process arrow, not equality. Page 31 keeps the addition statement and the reassembly statement distinct. Page 32 previews the two-label transition and recurrence of E03 without teaching its result or unlocking Level Three.

What the narrator explains after each game

After every mini-game, the narrator addresses the child directly. The short explanation says what the child did, what the result showed, the plain-language idea it teaches, and how that result advances the Moon Lantern mission. It follows an attempt and never supplies the solution in advance.

1. Find the whole lantern: all of the registered lantern is present, with no piece missing and no unrelated extra attached. That complete object is the whole the team must rebuild later.
2. Count the carrying spots: touching each of the four spots once gives the exact count. Skipping one or visiting one twice changes the record, so the child keeps the failed route visible and tries a new route.
3. Choose the four-part plan: the useful plan fills the round lantern frame with four separate, same-size pieces and no gap or overlap. The balanced left-right and top-bottom picture is a way to check the fit after looking.
4. Carry every lantern part: each of the same four visible lantern parts crosses the small doorway once. None is left behind and none is counted as a new extra part on the other side.
5. Compare the part sizes: Pair A stays level because its two parts are the same visible size. Pair B does not. Having the same number of pieces does not by itself show that the pieces are equal in size.
6. Check the gap and extra: a frame with a missing registered piece is not the complete lantern, and an unrelated extra does not become a lantern part merely because it is nearby. A complete rebuild needs every agreed part and nothing extra.
7. Name the held and whole counts: two selected pieces are held while four registered pieces make this whole plan. The labels matter because they tell the child what each count is counting.
8. Join the counts: each visible one-part group is used exactly once, so `1 + 1 + 1 + 1 = 4 parts`. Plus joins the counted groups, and equals says that both sides count the same four visible parts.
9. Build the Moon Lantern again: the four separate parts fit back into the one complete lantern the team saw at the start. `4 parts → 1 whole lantern` describes a rebuilding action; it does not say that the words four parts and one whole lantern are the same unit.

The final fourth-wall explanation should tell the child: one whole can be taken into agreed parts and rebuilt when every part is kept and counted once. The team checked for a missing part, a repeated part, a gap, an overlap and an extra before saying the lantern was complete. Addition helped join the four one-part counts into the same exact total. This matters when we build, pack, repair or check anything made from parts. The spoken version and caption must match exactly.

Safe optional materials

The book is complete without materials. Optional use may include large card circles cut by an adult, four large foam shapes, a fitted tray outline and four large floor markers. Avoid small loose parts, sharp cutting tools in the child's reach, glass, hot lamps and flashing lights. The child may watch rather than handle pieces.

Accessibility

- Keep words and reveal labels above pictures.
- Read picture descriptions aloud when helpful.
- Use shape, position, outline and text; never colour alone.
- Offer tactile outlines and large pieces with distinct edge textures.
- Allow extra processing time and unlimited replay without penalty.
- Keep candidate positions stable between challenge and reveal.
- Use the semantic HTML edition for scalable text and structured headings.
- The PDF will contain selectable text but will not be called fully tagged unless that is independently verified.

Companion Level Two requirements

Every paired stage must be directly playable by touch, keyboard and assistive technology and include a replay control. Mia, Sol and Tavi remain visible; Pax is the sole new Level Two guest. The final activity must require the child to place four visible parts, choose the total, read the equation and then rebuild the whole; a passive picture-memory sequence is not an acceptable substitute. The book-page bridge remains available after a correct answer. Optional codes unlock only character moments, alternate replay arrangements or an E03 preview.

Narration, including the recap and every post-game explanation, must be caption-matched and pre-rendered from Maria Smith's local Kokoro weights. Audio and sound effects are delivery tools only. The installed game must run offline, collect no personal or behavioural data and keep optional progress locally on the device.

Level Two has its own playful workshop theme, distinct from Level One's magical mystery music and Level Three's travelling rhythm. Gentle wooden taps, soft chimes or other building sounds may respond to play, but the background track must stay below speech and soften during narration. Provide a plainly labelled music control separate from narration and action sounds, and remember its setting on the device. Captions and visual changes must carry every necessary meaning when music is off. When the app loses focus, pause all sound; when it returns, do not restart a completed explanation or allow two clips or themes to overlap.

Connecting the book and Level Two

The book and game each give a complete route through the same lesson. The game adds physical manipulation, changing arrangements and replay; it must not alter the cause, count, vocabulary or final result established by the book.

- Start with the mission, not the answer: “The whole lantern cannot fit through the small door. How can we get the same lantern to the balcony?”
- Let the child inspect the complete lantern before showing separate parts, so they know what the parts must rebuild.
- Pair each book challenge with its matching mini-game in either order. After the child's attempt and the narrator's explanation, ask the child to point to the same evidence in the other format.
- Use process prompts such as “Have we used every part?”, “Did we visit that one already?”, “Is there a gap?” or “What are we counting?” Never point out the correct piece, route, total or position before an attempt.
- Keep a wrong arrangement visible long enough to compare it with the plan. Replay changes the puzzle arrangement, not the meaning of whole, part, plus or equals.
- Do not use a book code as a hint or prerequisite. It adds an optional moment, not an answer.

Reference-only curriculum navigation

The activities can support early communication, counting actions, shape comparison, spatial language, matching, fine-motor movement, recall and collaborative checking. UK early-years frameworks calibrate presentation only; they do not supply or validate any SFT premise.

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